

Software Requirements Specification (SRS)

Project Title

TalentBeacon™ – Employee Recommendation & Skill Intelligence Platform

Version

1.0

Project Domain

Data Science / Data Analytics

Project Duration

12–16 Weeks

Team Size

6–10 Interns

1. Introduction

1.1 Purpose

TalentBeacon™ is an AI-powered Employee Recommendation and Skill Intelligence Platform designed to identify the most suitable employees for specific job roles, projects, tasks, and career opportunities within an organization.

The platform analyzes employee skills, assessments, certifications, learning history, project experience, and performance metrics to:

- Recommend the best-fit employees for a role
 - Identify employees closest to a target role
 - Calculate skill gaps
 - Recommend courses and learning paths
 - Predict readiness for promotions and assignments
 - Support workforce planning and talent mobility
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1.2 Problem Statement

Organizations often face challenges such as:

- Difficulty identifying the right employee for a project
- Limited visibility into employee skills
- Lack of structured career progression
- Inability to identify skill gaps
- Manual talent allocation processes
- Poor workforce utilization

Managers frequently rely on personal knowledge instead of data-driven recommendations.

TalentBeacon™ solves these challenges through intelligent workforce analytics and recommendation algorithms.

1.3 Objectives

- Identify best-fit employees for job roles
 - Recommend internal candidates for projects
 - Measure employee-role similarity
 - Detect skill gaps
 - Recommend personalized learning paths
 - Improve workforce utilization
 - Support employee career growth
 - Enable data-driven talent decisions
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2. Scope

The platform shall provide:

- Employee Skill Repository
- Role Intelligence Engine
- Employee Recommendation Engine
- Skill Gap Analysis
- Learning Recommendation System
- Career Progression Analytics
- Workforce Readiness Analytics
- Employee Ranking Dashboard

The platform is intended for internal workforce development and talent management.

3. User Roles

Administrator

Responsibilities:

- Manage platform settings
 - Manage skills catalog
 - Manage learning resources
 - View organization-wide analytics
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Manager

Responsibilities:

- Search employees for projects
 - View recommendations
 - Analyze team skill gaps
 - Review employee readiness
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Employee

Responsibilities:

- View skill profile
 - View career recommendations
 - Access learning suggestions
 - Track role readiness
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4. Functional Requirements

Module 1: Authentication & Authorization

Features:

- Login
- Logout
- Password Reset
- Role-Based Access

Roles:

- Admin
 - Manager
 - Employee
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Module 2: Employee Skill Profile

Store employee information:

Personal Information

- Employee ID
- Name
- Department
- Designation

Professional Information

- Skills
 - Certifications
 - Learning Modules Completed
 - Assessments Completed
 - Project Experience
 - Performance Ratings
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Module 3: Skills Repository

Maintain master skill database.

Examples:

Technical Skills

- Python
- Java
- React
- AWS
- Docker
- SQL

Analytics Skills

- Power BI
- Tableau
- Statistics
- Excel

Soft Skills

- Communication
 - Leadership
 - Presentation
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Module 4: Role Intelligence Database

Store target job roles.

Examples:

- Full Stack Developer
 - Data Analyst
 - Data Scientist
 - Cloud Engineer
 - Cybersecurity Analyst
 - Machine Learning Engineer
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Role Profile Structure

Each role contains:

Required Skills

Example:

Data Analyst

- SQL
 - Excel
 - Power BI
 - Statistics
 - Python
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Desired Skills

- Machine Learning
 - Tableau
 - Data Visualization
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Minimum Skill Levels

Beginner

Intermediate

Advanced

Expert

Module 5: Employee Matching Engine

Purpose:

Identify employees closest to a target role.

Matching Parameters

Analyze:

- Skills
 - Certifications
 - Assessments
 - Learning History
 - Projects
 - Experience
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Match Score

Generate:

Employee Match Score

0% - 100%

Example:

Data Analyst Role

Employee A = 92%

Employee B = 81%

Employee C = 68%

Ranking

Display:

Top 10 Recommended Employees

Module 6: Skill Gap Analysis

Compare:

Employee Skills

vs

Target Role Skills

Example

Target Role:

Data Scientist

Required:

- Python
- Statistics
- Machine Learning
- SQL

Employee Skills:

- Python
- SQL

Gap:

- Statistics
 - Machine Learning
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Output

Skill Gap Report

Gap Severity Score

Module 7: Learning Recommendation Engine

Purpose:

Recommend courses to close skill gaps.

Recommendation Sources

Internal LMS

YouTube Learning Modules

Company Training Materials

External Certification Programs

Example

Target Role:

Cloud Engineer

Missing Skill:

AWS

Recommendations:

- AWS Fundamentals
 - EC2 Basics
 - IAM Fundamentals
 - VPC Networking
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Module 8: Career Path Recommendation

Suggest future career paths.

Example:

Current Role:

Python Developer

Potential Roles:

- Data Engineer
 - Machine Learning Engineer
 - Backend Engineer
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Output

Career Growth Roadmap

Estimated Readiness Percentage

Module 9: Workforce Readiness Score

Calculate:

Role Readiness Score

Based On:

- Skills
 - Assessments
 - Certifications
 - Experience
 - Learning Completion
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Example:

Data Scientist Readiness

82%

Module 10: Project Recommendation Engine

Input:

Project Requirements

Example:

Python

Flask

MySQL

AWS

Output:

Recommended Employees

Ranked by suitability

Module 11: Talent Discovery Dashboard

Manager can search:

- Skills
- Departments
- Roles
- Certifications

Example Queries:

Find employees skilled in AWS.

Find employees ready for promotion.

Find employees suitable for Data Scientist role.

Module 12: Workforce Analytics Dashboard

Display:

Organization Analytics

- Skill Distribution
- Skill Gaps
- Workforce Readiness

Learning Analytics

- Course Completion Rates
- Certification Trends

Talent Analytics

- Top Performers
 - Most Improved Employees
 - High Potential Employees
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Module 13: Employee Career Dashboard

Display:

Current Skills

Target Role

Match Percentage

Recommended Courses

Career Roadmap

Readiness Score

Learning Progress

Module 14: Reporting System

Generate Reports:

Employee Reports

- Skill Profile
- Readiness Score
- Learning Recommendations

Manager Reports

- Team Skill Gap Report
- Workforce Readiness Report

Organization Reports

- Talent Availability Report
- Skill Demand Report

Export:

- PDF
 - Excel
 - CSV
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5. Data Science Components

Employee Similarity Model

Calculate similarity between:

Employee Profile

and

Target Role Profile

Algorithms:

- Cosine Similarity
 - TF-IDF
 - Skill Vector Matching
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Recommendation Engine

Recommend:

- Employees
- Courses
- Career Paths

Methods:

- Content-Based Filtering
- Similarity Scoring

Readiness Prediction Model

Predict:

Role Readiness

Algorithms:

- Random Forest
- XGBoost
- Logistic Regression

Skill Gap Analysis Engine

Generate:

- Missing Skills
- Priority Skills
- Learning Recommendations

6. Non-Functional Requirements

Performance:

- Recommendation Generation < 5 Seconds
- Dashboard Load Time < 3 Seconds

Security:

- Role-Based Access
- Data Encryption
- Secure Authentication

Scalability:

- 10,000 Employees
- 5,000 Skills

- 1 Million Learning Records

Availability:

99% Uptime

7. Technology Stack

Frontend:

- HTML
- CSS
- Bootstrap
- JavaScript
- Chart.js

Backend:

- Python Flask

Database:

- MySQL

Data Analytics:

- Pandas
- NumPy

Machine Learning:

- Scikit-Learn
- XGBoost

Visualization:

- Plotly
- Power BI

Storage:

- AWS S3
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8. Project Milestones

Phase 1 (Weeks 1-3)

- Employee Skill Database
- Role Repository
- Authentication

Phase 2 (Weeks 4-6)

- Matching Engine
- Skill Gap Analysis

Phase 3 (Weeks 7-9)

- Learning Recommendation Engine
- Workforce Readiness Scoring

Phase 4 (Weeks 10-12)

- Dashboards
- Reports
- Analytics

Phase 5 (Weeks 13-16)

- ML Models
 - Testing
 - Documentation
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9. Expected Outcome

TalentBeacon™ will serve as an intelligent talent discovery and workforce recommendation platform capable of identifying the best employees for specific roles, measuring workforce readiness, detecting skill gaps, and generating personalized learning recommendations. The system will help organizations improve talent utilization, accelerate employee growth, and make data-driven workforce decisions.